ACADEMIC PORTFOLIO

CUONG MANH NGUYEN

Cuong Manh Nguyen (Louis)

Post & Telecommunication Institue of Technology (PTIT)
Ha Noi, Vietnam

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ABOUT ME

OBJECTIVE

Technology has always been more than just lines of code to me - it's a bridge to a better, more compassionate world. As a Computer Science graduate with a deep passion for AI, especially in speech processing, I'm driven by the belief that innovation should serve people, not just industries. Seeing the challenges faced by those struggling to access quality healthcare has fueled my commitment to building AI solutions that don't just promise change but deliver it, whether through early disease detection, personalized treatments, or more efficient patient care. I want to turn complex algorithms into real, tangible hope for those who need it most, ensuring that healthcare isn't just advanced but truly accessible to everyone.

EDUCATIONAL BACKGROUND

UNDERGRADUATE DEGREE

Post & Telecommunication Institute of Technology

Bachelor's in Information Technology

GPA 3.16/4.0

Bachelor thesis: Research to improve the quality of recommendation systems (Score: 10/10 - Top 0.01%)

08/2017 - 12/2021

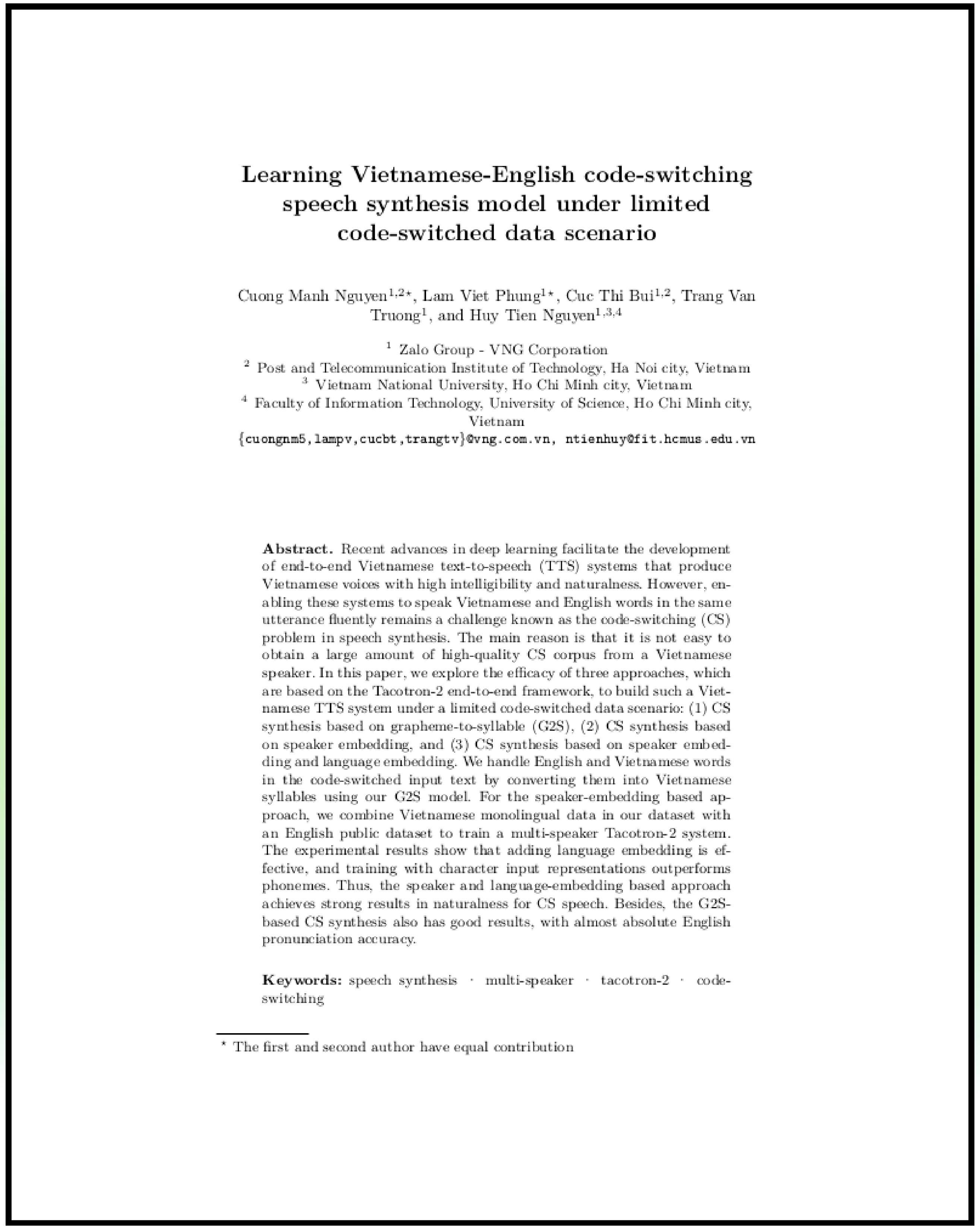
SCIENTIFIC RESEARCH

ABSTRACT

Recent advances in deep learning facilitate the development of end-to-end Vietnamese text-to-speech (TTS) systems that produce Vietnamese voices with high intelligibility and naturalness. However, en- abling these systems to speak Vietnamese and English words in the same utterance fluently remains a challenge known as the code-switching (CS) problem in speech synthesis. The main reason is that it is not easy to obtain a large amount of high-quality CS corpus from a Vietnamese speaker. In this paper, we explore the efficacy of three approaches, which are based on the Tacotron-2 end-to-end framework, to build such a Viet- namese TTS system under a limited code-switched data scenario: (1) CS synthesis based on grapheme-to-syllable (G2S), (2) CS synthesis based on speaker embedding, and (3) CS synthesis based on speaker embed- ding and language embedding. We handle English and Vietnamese words in the code-switched input text by converting them into Vietnamese syllables using our G2S model. For the speaker-embedding based ap- proach, we combine Vietnamese monolingual data in our dataset with an English public dataset to train a multi-speaker Tacotron-2 system. The experimental results show that adding language embedding is ef- fective, and training with character input representations outperforms phonemes. Thus, the speaker and language-embedding based approach achieves strong results in naturalness for CS speech. Besides, the G2S- based CS synthesis also has good results, with almost absolute English pronunciation accuracy.

“Learning Vietnamese-English code-switching speech synthesis model under limited code-switched data scenario“

Pacific Rim International Conference on Artificial Intelligence - Q2



Learning Vietnamese-English code-switching speech synthesis model under limited code-switched data scenario

Cuong Manh Nguyen[1,2*], Lam Viet Phung[1*], Cuc Thi Bui[1,2], Trang Van Truong[1], and Huy Tien Nguyen[1,3,4]

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[2] Post and Telecommunication Institute of Technology, Ha Noi city, Vietnam
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[4] Faculty of Information Technology, University of Science, Ho Chi Minh city, Vietnam

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Abstract. Recent advances in deep learning facilitate the development of end-to-end Vietnamese text-to-speech (TTS) systems that produce Vietnamese voices with high intelligibility and naturalness. However, enabling these systems to speak Vietnamese and English words in the same utterance fluently remains a challenge known as the code-switching (CS) problem in speech synthesis. The main reason is that it is not easy to obtain a large amount of high-quality CS corpus from a Vietnamese speaker. In this paper, we explore the efficacy of three approaches, which are based on the Tacotron-2 end-to-end framework, to build such a Vietnamese TTS system under a limited code-switched data scenario: (1) CS synthesis based on grapheme-to-syllable (G2S), (2) CS synthesis based on speaker embedding, and (3) CS synthesis based on speaker embedding and language embedding. We handle English and Vietnamese words in the code-switched input text by converting them into Vietnamese syllables using our G2S model. For the speaker-embedding based approach, we combine Vietnamese monolingual data in our dataset with an English public dataset to train a multi-speaker Tacotron-2 system. The experimental results show that adding language embedding is effective, and training with character input representations outperforms phonemes. Thus, the speaker and language-embedding based approach achieves strong results in naturalness for CS speech. Besides, the G2S-based CS synthesis also has good results, with almost absolute English pronunciation accuracy.

Keywords: speech synthesis · multi-speaker · tacotron-2 · code-switching

* The first and second author have equal contribution

SCIENTIFIC RESEARCH

ABSTRACT

News recommendation engines plays a vital role in online news services. Such systems help users discover news articles they may want to read, thus increase user experience and engagement. Existing news recommendation methods usually model users' interest from their reading history by learning user and news representations. Those meth- ods represent news by features specific to users to create personalized user interest models and recommend news similar to user interest. How- ever, in many cases users may also read news that are not in their usual interest, for example breaking news. Such a news often has global charac- teristics that are independent of specific users. In this paper, we propose a neural news recommendation method that combine user specific news models with global news models to better deal with those situations. To create informative news models, we adopt an attentive neural approach that can learn representations of news from different news aspects such as title, category, news contents. The attentive mechanism allows the model to focus on aspect that attract global interests from different users. Ex- periments on a real-world dataset show that our model achieves improved performance over state-of-the-art news recommendation method and is able to recommend candidate news of both types: those are relevant to a specific user and those are of common interest for different users.

"Combining User Specific and Global News Features for Neural News Recommendation "

Asian Conference on Intelligent Information and Database Systems - Q2

Combining User Specific and Global News Features for Neural News Recommendation

Cuong Manh Nguyen, Ngo Xuan Bach, and Tu Minh Phuong

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Abstract. News recommendation engines plays a vital role in online news services. Such systems help users discover news articles they may want to read, thus increase user experience and engagement. Existing news recommendation methods usually model users' interest from their reading history by learning user and news representations. Those methods represent news by features specific to users to create personalized user interest models and recommend news similar to user interest. However, in many cases users may also read news that are not in their usual interest, for example breaking news. Such a news often has global characteristics that are independent of specific users. In this paper, we propose a neural news recommendation method that combine user specific news models with global news models to better deal with those situations. To create informative news models, we adopt an attentive neural approach that can learn representations of news from different news aspects such as title, category, news contents. The attentive mechanism allows the model to focus on aspect that attract global interests from different users. Experiments on a real-world dataset show that our model achieves improved performance over state-of-the-art news recommendation method and is able to recommend candidate news of both types: those are relevant to a specific user and those are of common interest for different users.

Keywords: Recommender Systems · News Recommendation · Deep Neural Networks

1 Introduction

With the development of Internet, news readers have access to various sources of news online, which include online versions of traditional newspaper companies, or online news platforms such as Google News[1] and MSN News[2] that aggregate news from other news sites [1]. Such sources generate a very large number of news articles every day. Together with fast update cycle, such abundance of information makes it difficult for readers to find news that are most interesting for them. Personalized news recommendation techniques have shown to be useful

[1] https://news.google.com
[2] https://www.msn.com/en-us/news

SCIENTIFIC RESEARCH

ABSTRACT

An end-to-end text-to-speech (TTS) system (e.g. consisting of Tacotron-2 and WaveGlow vocoder) can achieve the state-of-the art quality in the presence of a large, professionallyrecorded training database. However, the drawbacks of using neural vocoders such as WaveGlow include 1) a time-consuming training process, 2) a slow inference speed, and 3) resource hunger when synthesizing waveform from spectral features. Moreover, the synthesized waveform from the neural vocoder can inherit the noise from an imperfect training data. This paper deals with the task of building Vietnamese TTS systems from moderate quality training data with noise. Our system utilizes an end-to-end TTS system that takes advantage of the Tacotron-2 acoustic model, and a custom vocoder combining a High Fidelity Generative Adversarial Networks (HiFiGAN)-based vocoder and a WaveGlow denoiser. Specifically, we used the HiFiGAN vocoder to achieve a better performance in terms of inference efficiency, and speech quality. Unlike previous works, we used WaveGlow as an effective denoiser to address the noisy synthesized speech. Moreover, the provided training data was thoroughly preprocessed using voice activity detection, automatic speech recognition and prosodic punctuation insertion. Our experiment showed that the proposed TTS system (as a combination of Tacotron-2, HiFiGAN-based vocoder, and WaveGlow denoiser) trained on the preprocessed data achieved a mean opinion score (MOS) of 3.77 compared to 4.22 for natural speech, which is the best result among participating systems of VLSP 2020's TTS evaluation.

"Development of Smartcall Vietnamese Text-to-Speech for VLSP 2020"

International Workshop on Vietnamese Language and Speech Processing

Development of Smartcall Vietnamese Text-to-Speech for VLSP 2020

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Smartcall JSC and ICTU, Thai Nguyen University

Abstract

An end-to-end text-to-speech (TTS) system (e.g. consisting of Tacotron-2 and WaveGlow vocoder) can achieve the state-of-the art quality in the presence of a large, professionally-recorded training database. However, the drawbacks of using neural vocoders such as WaveGlow include 1) a time-consuming training process, 2) a slow inference speed, and 3) resource hunger when synthesizing waveform from spectral features. Moreover, the synthesized waveform from the neural vocoder can inherit the noise from an imperfect training data. This paper deals with the task of building Vietnamese TTS systems from moderate quality training data with noise. Our system utilizes an end-to-end TTS system that takes advantage of the Tacotron-2 acoustic model, and a custom vocoder combining a High Fidelity Generative Adversarial Networks (HiFiGAN)-based vocoder and a WaveGlow denoiser. Specifically, we used the HiFiGAN vocoder to achieve a better performance in terms of inference efficiency, and speech quality. Unlike previous works, we used WaveGlow as an effective denoiser to address the noisy synthesized speech. Moreover, the provided training data was thoroughly preprocessed using voice activity detection, automatic speech recognition and prosodic punctuation insertion. Our experiment showed that the proposed TTS system (as a combination of Tacotron-2, HiFiGAN-based vocoder, and WaveGlow denoiser) trained on the preprocessed data achieved a mean opinion score (MOS) of 3.77 compared to 4.22 for natural speech, which is the best result among participating systems of VLSP 2020's TTS evaluation.

Index Terms— End-to-end TTS, Tacotron-2, HiFi-GAN, WaveGlow, vocoder

1 Introduction

Text-to-speech synthesis plays a crucial role in speech-based interaction systems. In the last two decades, there have been many attempts to build high quality Vietnamese TTS systems. A data processing scheme proved its efficacy in optimizing naturalness of end-to-end TTS systems trained on Vietnamese found data (Phung et al., 2020). Text normalization methods were explored; utilizing regular expressions and language model (Tuan et al., 2012). New prosodic features (e.g. phrase breaks) were investigated, which showed their efficacy in improving naturalness of Vietnamese hidden Markov models (HMM)-based TTS systems (Dinh et al., 2013; Trang et al., 2013; Phan et al., 2013). Different types of acoustic models were investigated such as HMM (Dinh et al., 2013), deep neural networks (DNN) (Nguyen et al., 2019), and sequence-to-sequence models (Phung et al., 2020). For postfiltering, it was shown that a global variance scaling method may destroy the tonal information; therefore, exemplar-based voice conversion methods were utilized in postfiltering to preserve the tonal information (Tuan et al., 2016). To our knowledge, there is little to none research on vocoders for Vietnamese TTS systems, especially when the training data is moderately noisy.

In the International Workshop on Vietnamese Language and Speech Processing (VLSP) 2020, a TTS challenge (Trang et al., 2020) required participants to build Vietnamese TTS systems from a provided moderately noisy corpus. The corpus included raw text and corresponding audio files. However, the corpus has incorrect pronunciation of a foreign language, the slight buzzer sounds in audio data, and many incorrectly labeled words, which pose significant challenges to participants. For example, a general neural vocoder will learn the buzzer sounds from the corpus, and introduce

LEADERSHIP ACHIEVEMENT

PRESIDENT OF PROGRAMMING PTIT CLUB

Programming PTIT (ProPTIT) Club is a dynamic professional environment. Members could participate in many programming courses from fundamental (e.g., C, C++, Java) to advanced (e.g., Data structures and algorithms, Games, Mobile apps, Web, Artificial Intelligence).

- Organized fundamental programming lectures and AI courses for students.
- Organized academic competitions and events that encouraged collaboration among club members.
- Managed club activities related to academics, communications, finance, and human resources.

Post & Telecommunication Institue of Technology, Ha Noi
12/2019 - 12/2020

GROUP LEADER
MID-AUTUMN FOR CHILDREN - LIGHTING UP VIETNAMESE CHILDHOOD

The "Mid-Autumn for Children – Lighting Up Vietnamese Childhood" is a meaningful volunteer blood donation program aimed at bringing hope and warmth to children in need, especially those facing medical challenges. This initiative not only encourages community engagement in blood donation but also spreads love and compassion during the Mid-Autumn Festival – a time traditionally dedicated to children.

- Directly advise and mobilize people to participate in voluntary blood donation.
- Managed team members and supported receiving blood donation applications on the official blood donation day.

TRUNG THU CHO EM
Thắp sáng tuổi thơ Việt

08/2017 - 11/2017

HORNORS ✦ AWARDS

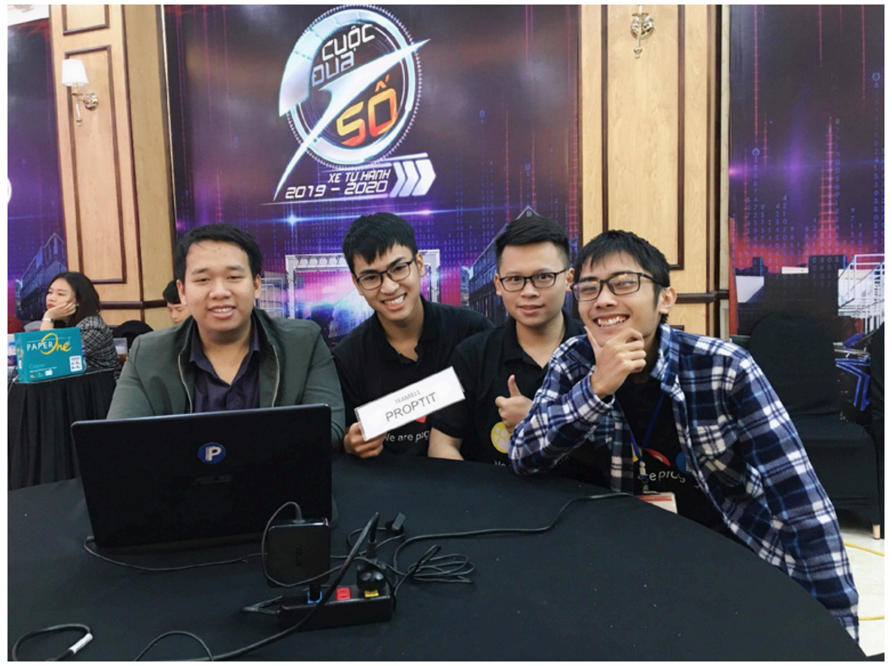
VIETNAMESE LANGUAGE AND SPEECH PROCESSING 2020

The Vietnamese Language and Speech Processing (VLSP) 2020 competition was a prestigious annual event focused on advancing natural language processing (NLP) and speech processing technologies for the Vietnamese language. Organized as part of the VLSP community's efforts, this competition aimed to bring together researchers, developers, and enthusiasts in the field of AI, machine learning, and computational linguistics.

FIRST PRIZE IN THE TEXT-TO-SPEECH CATEGORY

DIGITAL RACE 2020

The Cuộc Đua Số 2020, organized by FPT, was a prestigious autonomous vehicle programming competition for university students in Vietnam. The contest challenged participants to develop AI-driven self-driving car models capable of navigating complex road scenarios. It provided a platform for young tech enthusiasts to enhance their skills in AI, machine learning, and robotics.



Second prize - University round

Top 10 Finalists - National round

ACM/ICPC PTIT 2019

THIRD PRIZE

The ACM/ICPC PTIT 2019 (Association for Computing Machinery/International Collegiate Programming Contest - Posts and Telecommunications Institute of Technology) was a prestigious competitive programming contest held at PTIT (Posts and Telecommunications Institute of Technology) in Vietnam. This event was part of the global ACM ICPC series, known as the most challenging and competitive programming contest for university students worldwide.

Award: Top 6 over 174 teams

ACTIVITIES

VOLUNTEER

As a socially oriented person, here are some of my volunteer activities over the past year:

- Participate in supporting milk distribution for the elderly at the Ho Chi Minh City Oncology Hospital.
- Volunteer of the program "Cleaning House, Cleaning Heart."
- Accompanied the Nuoi Em non-profit organization to visit Vietnamese children in Cambodia.

Visit Ho Chi Minh City Oncology Hospital

Cleaning House, Cleaning Heart.

Visit Vietnamese children in Cambodia

RUNNING

Running is my passion, and nothing excites me more than tackling ultra-trail races. These incredible events allow me to fully immerse myself in the beauty of nature while pushing my limits to new heights. Whether navigating rugged terrains or savoring breathtaking landscapes, each race is an exhilarating journey that fuels my love for adventure and self-discovery.

CERTIFICATES

SOCIALIST REPUBLIC OF VIETNAM
Independence - Freedom - Happiness

THE DEGREE OF ENGINEER

INFORMATION TECHNOLOGY

POSTS AND TELECOMMUNICATIONS
INSTITUTE OF TECHNOLOGY
has conferred

Upon: Mr *Nguyen Manh Cuong*

Date of birth: 05 December 1999

Degree classification: Good

Given under the seal of
Posts and Telecommunications Institute of Technology

Reg. No: 0253/2022/DH

CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM
Độc lập - Tự do - Hạnh phúc

BẰNG KỸ SƯ

CÔNG NGHỆ THÔNG TIN

HỌC VIỆN CÔNG NGHỆ BƯU CHÍNH VIỄN THÔNG
cấp

Cho: Ông *Nguyễn Mạnh Cường*

Ngày sinh: 05 - 12 - 1999

Hạng tốt nghiệp: Khá

Hà Nội, ngày 18 tháng 03 năm 2022
GIÁM ĐỐC

HỌC VIỆN CÔNG NGHỆ BƯU CHÍNH VIỄN THÔNG
BỘ THÔNG TIN VÀ TRUYỀN THÔNG

PGS.TS Vũ Văn San

Số hiệu: D 01764

Số vào sổ gốc cấp văn bằng: 0253/2022/DH

IELTS™

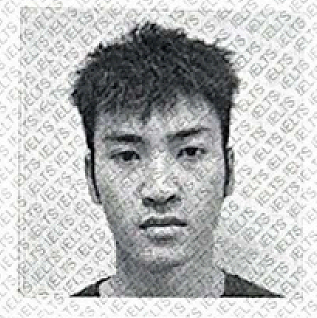
Test Report Form

ACADEMIC

NOTE *Admission to undergraduate and post graduate courses should be based on the ACADEMIC Reading and Writing Modules. GENERAL TRAINING Reading and Writing Modules are **not** designed to test the full range of language skills required for academic purposes. It is recommended that the candidate's language ability as indicated in this Test Report Form be re-assessed **after two years** from the date of the test.*

Centre Number: VN101 | Date: 23/MAR/2025 | Candidate Number: 532399

Candidate Details



Family Name: NGUYEN

First Name(s): MANH CUONG

Candidate ID: 030099007547

Date of Birth: 05/12/1999 | Sex (M/F): M | Scheme Code: Private Candidate

Country or Region of Origin:

Country of Nationality: VIET NAM

First Language: VIETNAMESE

Test Results

Listening	Reading	Writing	Speaking	Overall Band Score	CEFR Level
7.0	7.5	6.5	6.0	7.0	C1

Administrator Comments

Centre stamp

IDP Education idp VIETNAM

Validation stamp

VALIDATION STAMP IELTS

Administrator's Signature

Date: 24/03/2025

Test Report Form Number: 24VN532399NGUM101A

BRITISH COUNCIL idp CAMBRIDGE English

The validity of this IELTS Test Report Form can be verified online by recognising organisations at http://ielts.ucles.org.uk

DeepLearning.AI

COURSE CERTIFICATE

Jul 18, 2021

Cưởng Mạnh Nguyễn

has successfully completed

Neural Networks and Deep Learning

an online non-credit course authorized by DeepLearning.AI and offered through Coursera

Andrew Ng, Founder, DeepLearning.AI & Co-founder, Coursera
Kian Katanforoosh, Co-founder, Workera
Younes Bensouda Mourri, Instructor of AI, Stanford University

Verify at coursera.org/verify/7PQG43ZMF6G3

Coursera has confirmed the identity of this individual and their participation in the course.

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6 Courses

- Foundations of Project Management
- Project Initiation: Starting a Successful Project
- Project Planning: Putting It All Together
- Project Execution: Running the Project
- Agile Project Management
- Capstone: Applying Project Management in the Real World

Google

Apr 18, 2022

Cường Nguyễn Mạnh

has successfully completed the online, non-credit Professional Certificate

Google Project Management

Those who earn the Google Project Management Certificate have completed six courses, developed by Google, that include hands-on, practice-based assessments and are designed to prepare them for introductory-level roles in Project Management. They are competent in initiating, planning and running both traditional and agile projects.

The online specialization named in this certificate may draw on material from courses taught on-campus, but the included courses are not equivalent to on-campus courses. Participation in this online specialization does not constitute enrollment at this university. This certificate does not confer a University grade, course credit or degree, and it does not verify the identity of the learner.

Verify this certificate at:
https://coursera.org/verify/professional-cert/HTVWRYYFTLEU

ULTRA TRAIL
Nguyễn Mạnh Cường
67th
Overall
58th
Gender
11th
Category
10:44:05
Dalat Ultra Trail 2023
55 km
Timing Proivded by Sportstats

CAN THO
MARATHON
HERITAGE RACE
SINCE 2019
CERTIFICATION OF PARTICIPATION
AWARDED TO
Nguyễn Mạnh Cường
WHO FINISHED THE
FULL MARATHON 42.195 KM
AT THE CAN THO MARATHON - HERITAGE RACE
22 DECEMBER, 2024
IN A TOTAL TIME OF
03:58:45
GENDER: 154
OVERALL POSITION: 167
HEAD OF THE ORGANIZING COMMITTEE
CAN THO, 22 DECEMBER, 2024
ASSOC. PROF. NGUYEN TRI
GENERAL DIRECTOR, DHA VIETNAM

CAT TIEN
JUNGLE
PATHS
Race Jungle
CERTIFICATE
of completion
Congratulation!
NGUYỄN MẠNH CƯỜNG
has successfully completed the race
at CAT TIEN JUNGLE PATHS 2024 on December 15, 2024
DISTANCE
70KM
BIB
7005
TIME
10:52:24
OVERALL PLACE
14
GENDER PLACE
11
SPONSOR/PARTNER
Salonpas
COOL MATE
CÔNG TY CP ĐƯỜNG ĐUA MỚI
NEWRACE.,JSC
DIRECTOR
Đặng Quang Đức

THANK YOU